Frequency-Bottlenecked CycleGAN: Exposing and Patching Steganographic Vulnerabilities in Unpaired Medical Image Translation

Full Experimental Report (Internal)

CS424 Group Project — Task 2

April 2026

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1 Introduction

Unpaired image-to-image translation via CycleGAN enables translating between domains without paired training data. However, when the two domains are structurally asymmetric—such as brain MRI with tumors (pathological) versus healthy brain MRI—the cycle-consistency objective creates a perverse incentive. The forward generator $G : \mathcal{P} \to \mathcal{H}$ can minimise the cycle loss $\|F(G(p)) - p\|_1$ not by faithfully translating, but by **steganographically encoding** the tumor's location as imperceptible high-frequency noise in its output. The reverse generator F reads this hidden signal to reconstruct the original, achieving low cycle loss without honest translation.

We propose **FB-CycleGAN** (Frequency-Bottlenecked CycleGAN), which inserts a fixed, non-learnable Gaussian low-pass filter $B(\cdot)$ into the cycle-consistency path. This bottleneck destroys the high-frequency bandwidth required for steganographic encoding, forcing the generator to preserve pathology visibly or accept an irrecoverable cycle-consistency penalty.

This report presents a comprehensive forensic analysis of the baseline CycleGAN and FB-CycleGAN across five evaluation methods, establishing both the existence of steganography in the baseline and the effectiveness (and limitations) of the frequency bottleneck.

2 Background and Problem Formulation

2.1 CycleGAN Objective

CycleGAN learns two generators $G : \mathcal{P} \to \mathcal{H}$ and $F : \mathcal{H} \to \mathcal{P}$ with discriminators D_H and D_P. The full objective is:

$$\mathcal{L}_{\text{total}} = \mathcal{L}_{\text{GAN}}(G, D_H) + \mathcal{L}_{\text{GAN}}(F, D_P) + \lambda_{\text{cyc}}\mathcal{L}_{\text{cyc}}(G, F) + \lambda_{\text{id}}\mathcal{L}_{\text{id}}(G, F) \tag{1}$$

where the cycle-consistency loss enforces $F(G(p)) \approx p$ and $G(F(h)) \approx h$ via L1 distance, and the identity loss regularises generators to preserve in-domain inputs.

2.2 The Steganography Problem

When domains $\mathcal{P}$ and $\mathcal{H}$ are structurally asymmetric, G faces a dilemma. Honest translation (removing the tumor) makes cycle reconstruction impossible since F cannot know where the tumor was. Steganographic encoding (hiding tumor coordinates as invisible high-frequency patterns) achieves perfect cycle reconstruction without honest translation.

The steganographic channel capacity of a 256×256 image is vast—even 1 bit per pixel provides 65,536 bits, far more than needed to encode a tumor mask. Gradient descent converges to the steganographic solution because it achieves lower training loss.

2.3 FB-CycleGAN: The Frequency Bottleneck

We replace the standard cycle loss with:

$$\mathcal{L}_{\text{FB-cyc}}(G, F) = \mathbb{E}_p\left[\|F(B(G(p))) - p\|_1\right] + \mathbb{E}_h\left[\|G(B(F(h))) - h\|_1\right] \tag{2}$$

where $B(\cdot)$ is a Gaussian blur with fixed kernel size k and standard deviation σ. In the Fourier domain, B multiplies each frequency coefficient by $e^{-2\pi^2\sigma^2(u^2+v^2)}$, which is approximately 0 for frequencies beyond the cutoff $f_c = \frac{1}{2\pi\sigma}$.

The blur destroys the high-frequency subspace where steganographic data resides while preserving the low-frequency structural content. It adds zero learnable parameters and is fully differentiable, allowing gradient flow through the cycle path.

Critical implementation detail: The blur is applied exclusively in the cycle-consistency path—between G's output and F's input. It is *not* applied before the discriminator (which would change adversarial training dynamics), in the identity loss path, or at test-time inference.

3 Experimental Setup

3.1 Dataset: BraTS 2020

We use the BraTS 2020 training set containing 234 multi-parametric MRI volumes. We extract FLAIR (Fluid Attenuated Inversion Recovery) axial slices, chosen for their high tumor-to-background contrast.

Preprocessing pipeline:

1. Load NIfTI volumes with nibabel
2. Extract 2D axial slices from the FLAIR sequence
3. Classify each slice: **pathological** if tumor area $> 5\%$ of brain region, **healthy** if no tumor and brain area > 1000 pixels, otherwise skip
4. Center-crop to brain bounding box, resize to 256×256 via bilinear interpolation
5. Z-score normalise per volume (using non-zero voxel statistics), then clip to $[-3, 3]$ and scale to $[-1, 1]$

Dataset statistics:

	Train	Val	Test
Patients	187	23	24
Pathological slices	8,159	1,065	—
Healthy slices	12,169	1,467	—

Patient-level splitting ensures no data leakage between sets.

3.2 Architecture

Generator: ResNet-9 blocks. Encoder: ReflectionPad(3) $\rightarrow$ Conv(7, 64) $\rightarrow$ InstanceNorm $\rightarrow$ ReLU, two stride-2 downsampling convolutions, nine residual blocks, two transposed convolutions for upsampling, final Conv(7) $\rightarrow$ Tanh. Single-channel input/output (grayscale MRI). Parameters: 11.37M per generator.

Discriminator: 70×70 PatchGAN. Five convolutional layers with 4×4 kernels. Receptive field calculation: $4 \rightarrow 10 \rightarrow 22 \rightarrow 46 \rightarrow 70$. LSGAN (least-squares) objective. Parameters: 2.76M per discriminator.

Total model parameters: $2 \times 11.37\text{M} + 2 \times 2.76\text{M} = 28.26\text{M}$.

3.3 Training Configuration

Hyperparameter	Value
Optimiser	Adam ($\beta_1 = 0.5$, $\beta_2 = 0.999$)
Learning rate	2×10^{-4} (both G and D)
LR schedule	Constant for 100 epochs, linear decay to 0
Batch size	8
λ_{cyc}	10.0
λ_{id}	0.5 (effective weight: $0.5 \times 10 = 5$)
Replay buffer	50 images
Training epochs	100

3.4 Models Evaluated

Model	Bottleneck	σ	Kernel Size
Baseline CycleGAN	No	—	—
FB-CycleGAN $\sigma = 0.5$	Yes	0.5	3
FB-CycleGAN $\sigma = 1.0$	Yes	1.0	5
FB-CycleGAN $\sigma = 1.5$	Yes	1.5	7
FB-CycleGAN $\sigma = 2.0$	Yes	2.0	9

All models share identical architecture and hyperparameters; the only difference is the presence and strength of the Gaussian blur in the cycle path.

4 Hypotheses

Hypothesis 1 (H1: Steganographic Encoding)**.** The baseline CycleGAN exhibits structured high-frequency energy in FFT residuals and catastrophic reconstruction degradation under mild perturbation, confirming steganographic encoding.

Hypothesis 2 (H2: Bottleneck Effectiveness)**.** FB-CycleGAN shows diffuse, unstructured residual spectra and degrades gracefully under perturbation, confirming that the frequency bottleneck eliminates high-frequency steganographic encoding.

Hypothesis 3 (H3: Pareto-Optimal Operating Point)**.** There exists a σ^* achieving both acceptable generation quality (FID, SSIM) and strong steganography suppression.

Hypothesis 4 (H4: Structural Leakage)**.** Even with steganographic encoding eliminated, generators may retain visible structural traces of the source pathology that can be detected by a downstream classifier.

5 Evaluation Method 1: FFT Residual Analysis

5.1 Method

For each pathological input x and its generated "healthy" translation $\hat{h} = G(x)$, we compute the residual $r = |\hat{h} - x|$. We then decompose r into its frequency components via the 2D Discrete Fourier Transform:

$$\hat{r}[u, v] = \sum_{m=0}^{M-1} \sum_{n=0}^{N-1} r[m, n] \cdot e^{-i2\pi(um/M+vn/N)} \tag{3}$$

The power spectrum $P[u, v] = |\hat{r}[u, v]|^2$ reveals the frequency distribution of the residual. We apply a Hann window before the FFT to reduce spectral leakage, shift the zero-frequency component to the centre, and compute the radially averaged power spectrum:

$$P(\rho) = \frac{1}{|\{(u, v) : \sqrt{u^2 + v^2} = \rho\}|} \sum_{\sqrt{u^2+v^2}=\rho} P[u, v] \tag{4}$$

averaged over 100 validation samples.

5.2 Theoretical Justification

An honest generator modifies only the tumor region, producing a spatially localised residual. By the scaling property of the Fourier transform, spatially localised signals have broad, low-frequency spectra that decay as $P(\rho) \propto 1/\rho^{\alpha}$.

A steganographic generator spreads hidden data across all pixels as fine patterns. These patterns contribute significant energy at high frequencies, causing the radial power spectrum to plateau rather than decay.

5.3 Results

The baseline CycleGAN shows elevated high-frequency energy in the radial power spectrum compared to FB-CycleGAN variants. The 2D power spectrum of the baseline residual exhibits energy spread to the spectral edges, while FB-CycleGAN spectra show tighter concentration around the centre.

However, the spectral differences between models are modest—the radial curves are close. This motivates the need for more sensitive forensic methods (Sections 6 and 8).

6 Evaluation Method 2: Perturbation Robustness Test

6.1 Method

We inject additive Gaussian noise $\epsilon \sim \mathcal{N}(0, \sigma_\epsilon^2)$ into the generated intermediate image before feeding it to the reverse generator:

$$\hat{p}_{\text{perturbed}} = F(\hat{h} + \epsilon) \tag{5}$$

and measure the L1 reconstruction error $\|\hat{p}_{\text{perturbed}} - p\|_1$ as a function of $\sigma_\epsilon \in \{0, 0.001, 0.005, 0.01, 0.02, 0.05, 0.1\}$ averaged over 50 validation samples.

6.2 Theoretical Justification

In the Fourier basis, white noise has a flat power spectrum: $|\hat{\epsilon}[u, v]|^2 \approx \sigma_\epsilon^2$ for all (u, v). The steganographic signal $\hat{h}_{\text{hidden}}$ lives in the high-frequency subspace where $|\hat{h}_{\text{hidden}}[u, v]|$ is small. Even at $\sigma_\epsilon = 0.001$, the signal-to-noise ratio at high frequencies approaches 1, corrupting the hidden channel.

An honest generator's reconstruction relies on low-frequency structural features where $|\hat{h}_{\text{visible}}[u, v]| \gg |\hat{\epsilon}[u, v]|$, so noise has negligible effect.

6.3 Results

	L1 Reconstruction Error					Degradation
Model	$\sigma_\epsilon = 0$	0.001	0.005	0.01	0.1	Ratio
Baseline	0.051	0.122	0.207	0.209	0.265	**2.4×**
FB $\sigma = 0.5$	0.052	0.054	0.052	0.051	0.053	1.0×
FB $\sigma = 1.0$	0.054	0.053	0.051	0.050	0.054	1.0×
FB $\sigma = 1.5$	0.055	0.055	0.053	0.053	0.054	1.0×
FB $\sigma = 2.0$	0.058	0.058	0.057	0.056	0.058	1.0×

Key finding: The baseline shows a **2.4× catastrophic jump** in reconstruction error from $\sigma_\epsilon = 0$ to $\sigma_\epsilon = 0.001$—imperceptible noise causes massive reconstruction failure, proving dependence on a hidden high-frequency signal. All FB-CycleGAN variants maintain flat error curves regardless of noise level, confirming that reconstruction relies on visible structural features.

H1 confirmed: The baseline is steganographic. **H2 confirmed:** The frequency bottleneck eliminates high-frequency steganographic encoding at all tested σ values, including $\sigma = 0.5$.

7 Evaluation Method 3: Translation Quality Metrics

7.1 Method

We compute FID (Fréchet Inception Distance) and SSIM (Structural Similarity Index Measure) on 200 validation samples. For FID, we repeat the single grayscale channel to 3 channels for

Inception-v3 compatibility and resize to 299×299. SSIM is computed with data range 2.0 (images in $[-1, 1]$).

7.2 Results

Model	σ	$\text{SSIM}_{\text{A}\to\text{B}}$	SSIM_{cyc}	$\text{FID}_{\text{A}\to\text{B}}\downarrow$	$\text{FID}_{\text{B}\to\text{A}}\downarrow$
Baseline	—	0.289	0.777	101.3	125.1
FB $\sigma = 0.5$	0.5	0.292	**0.807**	105.6	133.1
FB $\sigma = 1.0$	1.0	0.283	0.787	107.5	125.2
FB $\sigma = 1.5$	1.5	0.292	0.779	104.4	135.4
FB $\sigma = 2.0$	2.0	**0.295**	0.771	**93.8**	**128.0**

Key findings:

1. **$\text{SSIM}_{\text{A}\to\text{B}}$ (translation quality)** is nearly identical across all models ($\sim$0.29), indicating the bottleneck does not degrade translation quality.
2. **SSIM_{cyc} (cycle reconstruction)** is highest for FB $\sigma = 0.5$ (0.807 vs baseline 0.777). The bottleneck forces visible structural preservation, which is more robust than steganographic encoding.
3. **$\text{FID}_{\text{A}\to\text{B}}$ improves with higher σ**: baseline 101.3 vs FB $\sigma = 2.0$ at 93.8. The bottleneck prevents the generator from wasting capacity on invisible encoding, focusing it instead on producing realistic translations.

H3 confirmed: All FB variants achieve comparable or better quality metrics than the baseline while eliminating steganography. The Pareto-optimal point depends on the metric prioritised: $\sigma = 0.5$ for cycle SSIM, $\sigma = 2.0$ for FID.

8 Evaluation Method 4: Classifier Leakage Test

8.1 Method

We train a lightweight ResNet-18 classifier on generated "healthy" images $\hat{h} = G(p)$ to predict properties of the original pathological input:

- **Tumor quadrant:** Which of 4 brain quadrants contains the most tumor mass (chance = 25%)
- **Tumor size:** Small, medium, or large based on area terciles (chance = 33%)

Tumor labels are extracted from the original BraTS segmentation masks. The classifier is trained for 20 epochs with Adam (lr = 10^{-3}) on 8,159 generated training images and evaluated on 1,065 validation images. We report best validation accuracy.

If the classifier achieves accuracy significantly above chance (threshold: 1.5$\times$ chance), the generated images contain recoverable information about the source pathology—evidence of information leakage.

8.2 Results

	Quadrant (chance = 25%)		Size (chance = 33%)	
Model	Best Val Acc	Verdict	Best Val Acc	Verdict
Baseline	66.9%	LEAKING	53.9%	LEAKING
FB $\sigma = 0.5$	76.0%	LEAKING	58.0%	LEAKING
FB $\sigma = 1.0$	74.1%	LEAKING	63.6%	LEAKING
FB $\sigma = 1.5$	77.1%	LEAKING	55.4%	LEAKING
FB $\sigma = 2.0$	79.2%	LEAKING	61.2%	LEAKING

Key finding: All models—including all FB-CycleGAN variants—leak tumor information to the classifier at rates far above chance. Notably, FB models show *higher* classifier accuracy than the baseline (76–79% vs 67%).

Interpretation: This result appears to contradict the perturbation test (which showed FB models are honest). The resolution lies in the *type* of leakage. The perturbation test detects invisible high-frequency encoding; the classifier detects *any* recoverable information, including visible structural traces. The FB bottleneck forces the generator to preserve tumor-related structural features visibly rather than hiding them, making the structural residue *more* classifiable even as the steganographic channel is destroyed.

This motivates the control experiments in Section 9.

9 Evaluation Method 5: Classifier Control Experiments

To distinguish steganographic leakage from structural leakage, and to rule out dataset bias, we conduct three control experiments.

9.1 Experiment 5a: Blur-then-Classify

Method: Apply Gaussian blur ($\sigma = 2.0$, $k = 9$) to the generated "healthy" images *before* feeding them to the classifier. This destroys any remaining high-frequency encoding while preserving visible structural features.

Prediction: If leakage is steganographic (high-frequency), blurring will reduce classifier accuracy. If leakage is structural (visible), blurring will have no effect.

Results:

Model	Clean Acc	Blurred Acc	Drop
Baseline	70.1%	72.3%	−2.2% (no drop)
FB $\sigma = 1.0$	75.2%	73.0%	+2.3% (no drop)

Finding: Blurring does not reduce classifier accuracy for either model. The leakage in both cases is **visible structural information**, not high-frequency steganographic encoding. Even the baseline's leakage to the classifier comes from structural cues, not from the steganographic channel that the perturbation test detected.

This means the perturbation test and classifier test detect *different phenomena*: the perturbation test detects the hidden channel used for *cycle reconstruction*, while the classifier detects visible structural residue that is inherent to imperfect translation.

9.2 Experiment 5b: Real Healthy Control (Dataset Bias)

Method: Train the same ResNet-18 classifier on *real* healthy brain images from the validation set, paired with *random* quadrant and size labels drawn from the pathological label distribution.

If the classifier achieves above-chance accuracy, there is inherent dataset bias (e.g., systematic brain asymmetry) rather than generator leakage.

Results:

Task	Real Healthy Acc	Chance
Quadrant	35.5%	25.0%
Size	37.2%	33.3%

Finding: Real healthy images with random labels give near-chance accuracy (35.5% vs 25% chance for quadrant). This is below the 1.5× threshold (37.5%), confirming **no significant dataset bias**. The 67–79% accuracy on generated images is genuine information leakage from the translation process, not a dataset artefact.

9.3 Experiment 5c: GradCAM Attention Maps

Method: We compute GradCAM (Gradient-weighted Class Activation Mapping) heatmaps for the trained classifiers, visualising which spatial regions the classifier attends to when predicting tumor quadrant. We compare attention patterns between classifiers trained on baseline vs FB-CycleGAN generated images.

Finding: Both classifiers focus attention on specific brain regions rather than diffuse areas, consistent with reading localised structural features. This further confirms that the leakage detected by the classifier is structural (localised to former tumor regions) rather than steganographic (spread across all pixels).

10 Pareto Analysis: Honesty vs Quality

We synthesise all metrics into a comprehensive Pareto analysis across σ values.

Model	σ	Degrad.	Cycle L1	FID↓	SSIM_{cyc}	Classifier
Baseline	—	2.4×	0.051	101.3	0.777	66.9%
FB $\sigma = 0.5$	0.5	1.0×	0.052	105.6	**0.807**	76.0%
FB $\sigma = 1.0$	1.0	1.0×	0.054	107.5	0.787	74.1%
FB $\sigma = 1.5$	1.5	1.0×	0.055	104.4	0.779	77.1%
FB $\sigma = 2.0$	2.0	1.0×	0.058	**93.8**	0.771	79.2%

Key observations:

1. **Even $\sigma = 0.5$ fully eliminates steganography** (degradation ratio drops from 2.4× to 1.0×). Stronger blurs provide no additional anti-steganographic benefit.
2. **FID improves with higher σ**: the bottleneck redirects generator capacity from hidden encoding to actual translation quality.
3. **Cycle SSIM peaks at $\sigma = 0.5$** then decreases, reflecting the tradeoff between blur strength and reconstruction fidelity.
4. **Classifier leakage increases with σ**: stronger blurs force more information into the visible structural domain.
5. **Recommended operating point:** $\sigma = 0.5$, which achieves steganography elimination, best cycle SSIM, and acceptable FID with minimal quality cost.

11 Discussion

11.1 Two Types of Information Leakage

Our experiments reveal that CycleGAN information leakage occurs through two distinct mechanisms:

1. **Steganographic leakage** (high-frequency encoding): The generator hides source information as invisible patterns in the translated image. Detected by the perturbation test (catastrophic degradation under noise) and FFT analysis (elevated high-frequency power). *Eliminated by the frequency bottleneck.*

2. **Structural leakage** (visible residue): The generator fails to perfectly erase all visible traces of the source pathology, leaving structural artefacts (tissue distortion, intensity asymmetry). Detected by the classifier leakage test but *not* by the perturbation test. *Not eliminated by the frequency bottleneck.*

This distinction is critical. Prior work on CycleGAN steganography focused exclusively on invisible encoding. Our classifier leakage test reveals a second, orthogonal failure mode that persists even after the steganographic channel is destroyed.

11.2 Why the Bottleneck Improves FID

The counterintuitive improvement in FID with the bottleneck (101.3 $\rightarrow$ 93.8 at $\sigma = 2.0$) has a clear explanation in the Fourier framework. Without the bottleneck, the generator allocates capacity to two tasks: (1) producing a realistic-looking translation, and (2) encoding hidden information for cycle reconstruction. The bottleneck makes task (2) impossible, so the generator's full capacity is directed toward task (1). The result is a more realistic translation despite—or rather because of—the constraint.

This is analogous to regularisation in supervised learning: removing a degree of freedom (the steganographic channel) can improve generalisation by preventing the model from exploiting a shortcut.

11.3 Limitations

1. **Single seed:** All experiments use a single random seed. Multiple seeds with statistical testing would strengthen the claims.

2. **No downstream segmentation:** The proposal included U-Net evaluation (Dice score on synthetic data). This remains as future work.

3. **Structural leakage unsolved:** The bottleneck addresses steganographic encoding but not visible structural residue. An adversarial classifier head during training could potentially reduce this.

4. **Blur type not ablated:** We tested only Gaussian blur. Other bottlenecks (median filter, box blur, learned low-pass) may offer different tradeoffs.

12 Conclusion

We presented FB-CycleGAN, a zero-parameter frequency bottleneck that eliminates steganographic encoding in unpaired medical image translation. Through five complementary evaluation methods—FFT spectral analysis, perturbation robustness testing, FID/SSIM quality metrics, classifier leakage testing, and control experiments—we established:

1. **The baseline CycleGAN is steganographic:** The perturbation test shows 2.4× catastrophic reconstruction degradation from imperceptible noise ($\sigma_\epsilon = 0.001$).
2. **The frequency bottleneck eliminates steganographic encoding:** All FB-CycleGAN variants show flat perturbation curves (1.0× degradation ratio) at every tested σ.
3. **Translation quality is preserved or improved:** SSIM is comparable across models; FID improves from 101.3 (baseline) to 93.8 (FB $\sigma = 2.0$).
4. **Structural leakage persists:** A classifier can recover tumor properties from generated "healthy" images at 67–79% accuracy (chance = 25%) for all models. Control experiments confirm this is visible structural residue, not steganographic encoding.
5. $\sigma = 0.5$ **is sufficient:** Even the lightest blur fully eliminates steganography while maximising cycle reconstruction fidelity.

The distinction between steganographic and structural leakage—revealed by combining the perturbation test with the classifier leakage test and the blur-then-classify control—is, to our knowledge, a novel analytical contribution. It suggests that future work on honest unpaired translation must address both the invisible encoding channel and the visible translation quality.